

Foundations of Sigmoid function

[Sigmoid function]

1.0 Content Block

A sigmoid function is any mathematical function whose graph has a characteristic S-shaped or sigmoid curve. A common example of a sigmoid function is the logistic function. Other sigmoid functions are given in the Examples section. In some fields, most notably in the context of artificial neural networks, the term "sigmoid function" is used as a synonym for "logistic function". Special cases of sigmoid functions include the Gompertz curve (used in modeling systems that saturate at large values of x) and the ogee curve (used in the spillway of some dams). Sigmoid functions have domain of all real numbers, with return (response) value commonly monotonically increasing but could be decreasing. Sigmoid functions most often show a return value (y axis) in the range 0 to 1. Another commonly used range is from -1 to 1 . There is also the Heaviside step function, which instantaneously transitions between 0 and 1. A wide variety of sigmoid functions including the logistic and hyperbolic tangent functions have been used as the activation function of artificial neurons. Sigmoid curves are also common in statistics as cumulative distribution functions (which go from 0 to 1), such as the integrals of the logistic density, the normal density, and Student's t probability density functions. The logistic sigmoid function is invertible, and its inverse is the logit function.

2.0 Content Block

Hyperbolic tangent (shifted and scaled version of the logistic function, above) $f(x) = \tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

3.0 Content Block

Gudermannian function $f(x) = \operatorname{gd}(x) = \int_0^x \frac{dt}{\cosh t} = 2 \arctan \left(\tanh \left(\frac{x}{2} \right) \right)$

4.0 Content Block

1st kind A unitary sigmoid function of the first kind is a bounded increasing function that approaches its lower and upper asymptotes monotonically, without oscillation. This class includes many of the standard sigmoid functions used in statistics, biomathematics, and engineering, such as the logistic function and related generalizations.

5.0 Content Block

M0: Construction method for sigmoid functions not evident or intuitive M1: Inverse of singularity functions M2: Sigmoid functions of embedded positive functions M3: Rising a sigmoid function to the power M4: Exponentiating a sigmoid function M5: Symmetric sigmoid functions derived from asymmetric ones M6: Sigmoid functions of the reciprocal independent variable M7: Embedding a sigmoid function into other function M8: Sum of sigmoid functions M9: Multiplication of sigmoid functions M10: Integral of the product of an increasing and a decreasing function M11: Derivation from lambda (bell-shaped) functions M12: Integration of lambda (bell-shaped) function M13: Integration of the sum of lambda (bell-shaped) functions M14: Integration of the product of two lambda (bell-shaped) functions M15: Integration of the difference of two shifted sigmoid functions M16: Integration of the product of two shifted sigmoid functions M17: Convolution of sigmoid functions M18: Integration of the product of lambda and sigmoid function M19: Solutions of ordinary differential equations M20: Solutions of partial differential equation (PDE) M21: Solutions of functional differential equation (FDE) M22: Sum of a sigmoid function and some derivatives

M23: Combination of sigmoid functions, its derivative and integral M24: Filtering sigmoid functions M25: Special cases of Gauss hypergeometric functions M26: Feedback closed-loop systems M27: Recursive functions M28: Recursive time-delayed feed-forward loops M29: Solutions of stochastic differential equation M30: Chaotic sigmoid functions Consult reference for more details.

6.0 Content Block

Smoothstep function $f(x) = \begin{cases} 0 & |x| \leq 1 \\ \frac{3}{4}x^2 - \frac{1}{2}x^3 & |x| < 2 \\ 1 & |x| \geq 2 \end{cases}$ $\{ \displaystyle f(x) = \begin{cases} 0 & |x| \leq 1 \\ \frac{3}{4}x^2 - \frac{1}{2}x^3 & |x| < 2 \\ 1 & |x| \geq 2 \end{cases} \quad N \in \mathbb{Z} \geq 1 \}$

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Up to shifts and scaling, many sigmoids are special cases of $f(x) = \phi(\phi(x, \beta), \alpha)$, $\{ \displaystyle f(x) = \varphi(\varphi(x, \beta), \alpha) \}$ where $\phi(x, \lambda) = \begin{cases} (1 - \lambda x)^{1/\lambda} & \lambda \neq 0 \\ e^{-x} & \lambda = 0 \end{cases}$ $\{ \displaystyle \varphi(x, \lambda) = \begin{cases} (1 - \lambda x)^{1/\lambda} & \lambda \neq 0 \\ e^{-x} & \lambda = 0 \end{cases} \}$ is the inverse of the negative Box-Cox transformation, and $\alpha < 1$ $\{ \displaystyle \alpha < 1 \}$ and $\beta < 1$ $\{ \displaystyle \beta < 1 \}$ are shape parameters. Smooth transition function normalized to $(-1, 1)$:

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Further reading Mitchell, Tom M. (1997). Machine Learning. WCB McGraw-Hill. ISBN 978-0-07-042807-2.. (NB. In particular see "Chapter 4: Artificial Neural Networks" (in particular pp. 96–97) where Mitchell uses the word "logistic function" and the "sigmoid function" synonymously – this function he also calls the "squashing function" – and the sigmoid (aka logistic) function is used to compress the outputs of the "neurons" in multi-layer neural nets.) Humphrys, Mark. "Continuous output, the sigmoid function". Archived from the original on 2022-07-14. Retrieved 2022-07-14. (NB. Properties of the sigmoid, including how it can shift along axes and how its domain may be transformed.)

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Error function $f(x) = \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$ $\{ \displaystyle f(x) = \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \}$

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3rd kind A unitary sigmoid function of the third kind is a bounded increasing function that oscillates near both the lower and upper asymptotes. These functions retain the global shape of a sigmoid curve but exhibit oscillatory behavior in the vicinity of both limiting states.

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$f(x) = \begin{cases} \frac{2}{1 + e^{-2mx}} & |x| < 1 \\ \operatorname{sgn}(x) & |x| \geq 1 \end{cases} = \begin{cases} \tanh\left(\frac{mx}{1 - x^2}\right) & |x| < 1 \\ \operatorname{sgn}(x) & |x| \geq 1 \end{cases}$ $\{ \displaystyle \begin{aligned} f(x) &= \begin{cases} \frac{2}{1 + e^{-2mx}} & |x| < 1 \\ \operatorname{sgn}(x) & |x| \geq 1 \end{cases} \\ &= \begin{cases} \tanh\left(\frac{mx}{1 - x^2}\right) & |x| < 1 \\ \operatorname{sgn}(x) & |x| \geq 1 \end{cases} \end{aligned} \}$ using the hyperbolic tangent mentioned above. Here, m $\{ \displaystyle m \}$ is a free parameter encoding the slope at $x = 0$ $\{ \displaystyle x = 0 \}$, which must be greater than or equal to $\sqrt{3}$ $\{ \displaystyle \sqrt{3} \}$ because any smaller value will result in a function with multiple inflection points, which is therefore not a true sigmoid. This function is unusual because it actually attains the limiting values of -1 and 1 within a finite range, meaning that its value is constant at -1 for all $x \leq -1$ $\{ \displaystyle x \leq -1 \}$ and at 1 for all $x \geq 1$ $\{ \displaystyle x \geq 1 \}$. Nonetheless, it is smooth (infinitely differentiable, C^∞ $\{ \displaystyle C^\infty \}$) everywhere, including at $x = \pm 1$ $\{ \displaystyle x = \pm 1 \}$.